Contribution Journal – Module 9: Ensemble Methods

Student: John'a Matthews

1. Introduction

In Module 9, I learned about ensemble methods and how combining multiple machine learning models can improve predictions. The purpose of this lab was to compare a single Decision Tree with a Random Forest and understand how using multiple decision trees can help make predictions. I also learned how to identify which features are most important when classifying data.

2. Contribution

For this lab, I completed the coding activities using Google Colab and Python. I loaded the Iris dataset and prepared the data by splitting it into training and testing sets. I created and trained a single Decision Tree model and used it to predict the different Iris flower classes. I then created a Random Forest model with 100 decision trees and evaluated its accuracy.

I compared the results from both models and found that they each achieved 100% accuracy. I also completed the feature importance activity by using Matplotlib and NumPy to create a bar chart showing which flower measurements were most important. My results showed that petal length and petal width were the two most important features.

I answered the knowledge check questions and reviewed what I learned about the differences between Decision Trees and Random Forests. I made sure to understand the code and what each step was doing instead of just running it.

3. Conclusion

Overall, my contribution to Module 9 was completing the coding exercises, training and comparing both machine learning models, creating the feature importance chart, and answering the knowledge check questions. One thing I learned is that using more decision trees does not always guarantee higher accuracy. Both models performed equally well on the Iris dataset. This lab helped me better understand ensemble learning and gave me more experience using Python and Scikit-learn to train and evaluate machine learning models.